1) Develop a Shopping cart in PHP with following requirements:

Admin Site:
1) Login
2) Manage Category (CRUD)
3) Manage Products (Product data with pictures. Upload picture and resize them)
4) Logout

User Site:

Categories Garments Mobiles Laptops	Pic pname price p4	p2	p3

1) List Categories
2) Display products of category whose hyperlink is clicked(Apply pagination, load data with ajax).
3) Register user (Use CAPTCHA)
4) Login
---
2) Read data from students.xml file and store them in students table of MySQL.

Register.php

```php
<?php
session_start();
require_once 'db.php';

$error = '';
$success = '';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $captcha = trim($_POST['captcha'] ?? '');

    if (empty($captcha) || strcasecmp($captcha, $_SESSION['captcha'] ??
'') !== 0) {
        $error = "Incorrect CAPTCHA code!";
    } else {
        $username = trim($_POST['username']);
        $email = trim($_POST['email']);
        $password = password_hash($_POST['password'], PASSWORD_DEFAULT);

        $stmt = $conn->prepare("INSERT INTO users (username, email,
password) VALUES (?, ?, ?)");
        $stmt->bind_param("sss", $username, $email, $password);
        if ($stmt->execute()) {
```

```
            $success = "Registration successful! <a
href='user_login.php'>Login here</a>";
        } else {
            $error = "Username or Email already exists.";
        }
    }
}
?>
<!DOCTYPE html>
<html>
<head><title>User Registration</title></head>
<body>
    <h2>User Registration</h2>
    <?php
    if ($error) echo "<p style='color:red;'>$error</p>";
    if ($success) echo "<p style='color:green;'>$success</p>";
    ?>
    <form method="POST">
        <label>Username:</label><br>
        <input type="text" name="username" required><br><br>

        <label>Email:</label><br>
        <input type="email" name="email" required><br><br>

        <label>Password:</label><br>
        <input type="password" name="password" required><br><br>

        <label>CAPTCHA:</label><br>
        <img src="captcha.php" alt="CAPTCHA" style="vertical-align:middle;
margin-bottom:5px;"><br>
        <input type="text" name="captcha" placeholder="Enter CAPTCHA"
required><br><br>

        <button type="submit">Register</button>
    </form>
    <p>Already registered? <a href="user_login.php">Login here</a></p>
</body>
</html>
```

Login.php

```php
<?php
session_start();
require_once 'db.php';

$error = '';
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $username = trim($_POST['username']);
    $password = $_POST['password'];

    $stmt = $conn->prepare("SELECT id, password FROM users WHERE username
= ?");
    $stmt->bind_param("s", $username);
    $stmt->execute();
    $res = $stmt->get_result();

    if ($row = $res->fetch_assoc()) {
        if (password_verify($password, $row['password'])) {
            $_SESSION['user_logged'] = true;
            $_SESSION['user_name'] = $username;
            header("Location: index.php");
            exit;
        }
    }
    $error = "Invalid username or password.";
}
?>
<!DOCTYPE html>
<html>
<head><title>User Login</title></head>
<body>
    <h2>User Login</h2>
    <?php if($error) echo "<p style='color:red;'>$error</p>"; ?>
    <form method="POST">
        <label>Username:</label><br>
        <input type="text" name="username" required><br><br>
        <label>Password:</label><br>
        <input type="password" name="password" required><br><br>
        <button type="submit">Login</button>
    </form>
    <p>New user? <a href="register.php">Register here</a></p>
```

```
</body>
</html>
```

Index.php

```
<?php
session_start();
require_once 'db.php';

$categories = $conn->query("SELECT * FROM categories ORDER BY name ASC");
?>
<!DOCTYPE html>
<html>
<head>
    <title>Shopping Cart</title>
    <style>
        .layout-table { width: 100%; border-collapse: collapse;
margin-top: 15px; }
        .layout-table td { border: 1px solid #000; padding: 15px;
vertical-align: top; }
        .cat-sidebar { width: 25%; }
        .product-area { width: 75%; }
        ul { list-style-type: none; padding-left: 0; }
        ul li { margin-bottom: 8px; }
    </style>
</head>
<body>
    <div style="display:flex; justify-content:space-between;
align-items:center;">
        <h1>Shop Online</h1>
        <div>
            <?php if (isset($_SESSION['user_logged'])): ?>
                Logged in as <b><?=
htmlspecialchars($_SESSION['user_name']) ?></b> | <a
href="logout.php">Logout</a>
            <?php else: ?>
                <a href="user_login.php">Login</a> | <a
href="register.php">Register</a>
            <?php endif; ?>
        </div>
```

```
    </div>

     <table class="layout-table">
        <tr>
            <td class="cat-sidebar">
                <h3>Categories</h3>
                <ul>
                    <?php while ($c = $categories->fetch_assoc()): ?>
                        <li>
                            <a href="javascript:void(0);"
onclick="loadProducts(<?= $c['id'] ?>, 1)">
                                <?= htmlspecialchars($c['name']) ?>
                            </a>
                        </li>
                    <?php endwhile; ?>
                </ul>
            </td>
            <td class="product-area">
                <div id="product-container">
                    <i>Click on any category on the left to view
products.</i>
                </div>
            </td>
        </tr>
    </table>

    <script>
        function loadProducts(catId, page) {
            const container =
document.getElementById('product-container');
            container.innerHTML = 'Loading products...';

            const xhr = new XMLHttpRequest();
            xhr.open('GET', 'fetch_products.php?cat_id=' + catId +
'&page=' + page, true);
            xhr.onload = function() {
                if (this.status === 200) {
                    container.innerHTML = this.responseText;
                } else {
                    container.innerHTML = 'Failed to load products.';
```

```
                }
            };
            xhr.send();
        }
    </script>
</body>
</html>
```

Admin login.php

```
<?php
session_start();
require_once 'db.php';

$error = '';
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $username = trim($_POST['username']);
    $password = $_POST['password'];

    $stmt = $conn->prepare("SELECT id, password FROM admins WHERE username
= ?");
    $stmt->bind_param("s", $username);
    $stmt->execute();
    $res = $stmt->get_result();

    if ($row = $res->fetch_assoc()) {
```

```
        if (password_verify($password, $row['password'])) {
            $_SESSION['admin_logged'] = true;
            $_SESSION['admin_user'] = $username;
            header("Location: admin_dashboard.php");
            exit;
        }
    }
    $error = "Invalid username or password.";
}
?>
<!DOCTYPE html>
<html>
<head><title>Admin Login</title></head>
<body>
    <h2>Admin Login</h2>
    <?php if($error) echo "<p style='color:red;'>$error</p>"; ?>
    <form method="POST">
        <label>Username:</label><br>
        <input type="text" name="username" required><br><br>
        <label>Password:</label><br>
        <input type="password" name="password" required><br><br>
        <button type="submit">Login</button>
    </form>
</body>
</html>
```

Manage_categories.php

```
<?php
session_start();
if (!isset($_SESSION['admin_logged'])) {
    header("Location: admin_login.php");
    exit;
}
require_once 'db.php';

 if (isset($_POST['add_category'])) {
```

```
    $cname = trim($_POST['category_name']);
    if ($cname != '') {
        $stmt = $conn->prepare("INSERT INTO categories (name) VALUES
(?)");
        $stmt->bind_param("s", $cname);
        $stmt->execute();
    }
    header("Location: manage_categories.php");
    exit;
}

 if (isset($_POST['update_category'])) {
    $cid = intval($_POST['cat_id']);
    $cname = trim($_POST['category_name']);
    $stmt = $conn->prepare("UPDATE categories SET name = ? WHERE id = ?");
    $stmt->bind_param("si", $cname, $cid);
    $stmt->execute();
    header("Location: manage_categories.php");
    exit;
}

 if (isset($_GET['delete'])) {
    $cid = intval($_GET['delete']);
    $stmt = $conn->prepare("DELETE FROM categories WHERE id = ?");
    $stmt->bind_param("i", $cid);
    $stmt->execute();
    header("Location: manage_categories.php");
    exit;
}

 $edit_row = null;
if (isset($_GET['edit'])) {
    $cid = intval($_GET['edit']);
    $res = $conn->query("SELECT * FROM categories WHERE id = $cid");
    $edit_row = $res->fetch_assoc();
}

$categories = $conn->query("SELECT * FROM categories ORDER BY id DESC");
?>
<!DOCTYPE html>
```

```
<html>
<head><title>Manage Categories</title></head>
<body>
    <p><a href="admin_dashboard.php">&larr; Dashboard</a></p>
    <h2><?= $edit_row ? 'Edit Category' : 'Add Category' ?></h2>
    <form method="POST">
        <input type="hidden" name="cat_id" value="<?= $edit_row['id'] ??
'' ?>">
        <input type="text" name="category_name" required value="<?=
htmlspecialchars($edit_row['name'] ?? '') ?>">
        <button type="submit" name="<?= $edit_row ? 'update_category' :
'add_category' ?>">
            <?= $edit_row ? 'Update' : 'Add' ?>
        </button>
    </form>

    <h2>All Categories</h2>
    <table border="1" cellpadding="8" cellspacing="0">
        <tr><th>ID</th><th>Name</th><th>Action</th></tr>
        <?php while ($cat = $categories->fetch_assoc()): ?>
            <tr>
                <td><?= $cat['id'] ?></td>
                <td><?= htmlspecialchars($cat['name']) ?></td>
                <td>
                    <a href="?edit=<?= $cat['id'] ?>">Edit</a> |
                    <a href="?delete=<?= $cat['id'] ?>" onclick="return
confirm('Are you sure?')">Delete</a>
                </td>
            </tr>
        <?php endwhile; ?>
    </table>
</body>
</html>
```

Manage_products.php

```
<?php
```

```php
session_start();
if (!isset($_SESSION['admin_logged'])) {
    header("Location: admin_login.php");
    exit;
}
require_once 'db.php';

 function resizeAndSaveImage($sourcePath, $targetPath, $maxW = 300, $maxH
= 300) {
    list($origW, $origH, $type) = getimagesize($sourcePath);
    $ratio = min($maxW / $origW, $maxH / $origH);
    $newW = round($origW * $ratio);
    $newH = round($origH * $ratio);

    $thumb = imagecreatetruecolor($newW, $newH);

    switch ($type) {
        case IMAGETYPE_JPEG:
            $source = imagecreatefromjpeg($sourcePath);
            break;
        case IMAGETYPE_PNG:
            $source = imagecreatefrompng($sourcePath);
            imagealphablending($thumb, false);
            imagesavealpha($thumb, true);
            break;
        default:
            return false;
    }

    imagecopyresampled($thumb, $source, 0, 0, 0, 0, $newW, $newH, $origW,
$origH);

    if ($type == IMAGETYPE_PNG) {
        imagepng($thumb, $targetPath, 8);
    } else {
        imagejpeg($thumb, $targetPath, 85);
    }

    imagedestroy($thumb);
    imagedestroy($source);
```

```
    return true;
}

 if (isset($_POST['add_product'])) {
    $cat_id = intval($_POST['category_id']);
    $name = trim($_POST['name']);
    $price = floatval($_POST['price']);

    if (!empty($_FILES['image']['tmp_name'])) {
        if (!is_dir('uploads')) {
            mkdir('uploads', 0777, true);
        }
        $filename = time() . '_' . basename($_FILES['image']['name']);
        $targetFile = 'uploads/' . $filename;

        if (resizeAndSaveImage($_FILES['image']['tmp_name'], $targetFile,
250, 250)) {
            $stmt = $conn->prepare("INSERT INTO products (category_id,
name, price, image) VALUES (?, ?, ?, ?)");
            $stmt->bind_param("isds", $cat_id, $name, $price,
$targetFile);
            $stmt->execute();
        }
    }
    header("Location: manage_products.php");
    exit;
}

 if (isset($_GET['delete'])) {
    $pid = intval($_GET['delete']);
    $res = $conn->query("SELECT image FROM products WHERE id = $pid");
    if ($row = $res->fetch_assoc()) {
        if (file_exists($row['image'])) unlink($row['image']);
    }
    $conn->query("DELETE FROM products WHERE id = $pid");
    header("Location: manage_products.php");
    exit;
}

$categories = $conn->query("SELECT * FROM categories");
```

```
$products = $conn->query("SELECT p.*, c.name as category_name FROM
products p JOIN categories c ON p.category_id = c.id ORDER BY p.id DESC");
?>
<!DOCTYPE html>
<html>
<head><title>Manage Products</title></head>
<body>
    <p><a href="admin_dashboard.php">&larr; Dashboard</a></p>
    <h2>Add New Product</h2>
    <form method="POST" enctype="multipart/form-data">
        <label>Category:</label><br>
        <select name="category_id" required>
            <?php while($c = $categories->fetch_assoc()): ?>
                <option value="<?= $c['id'] ?>"><?=
htmlspecialchars($c['name']) ?></option>
            <?php endwhile; ?>
        </select><br><br>

        <label>Product Name:</label><br>
        <input type="text" name="name" required><br><br>

        <label>Price:</label><br>
        <input type="number" step="0.01" name="price" required><br><br>

        <label>Product Image (JPG/PNG):</label><br>
        <input type="file" name="image" accept="image/jpeg,image/png"
required><br><br>

        <button type="submit" name="add_product">Save Product</button>
    </form>

    <h2>Product List</h2>
    <table border="1" cellpadding="8" cellspacing="0">

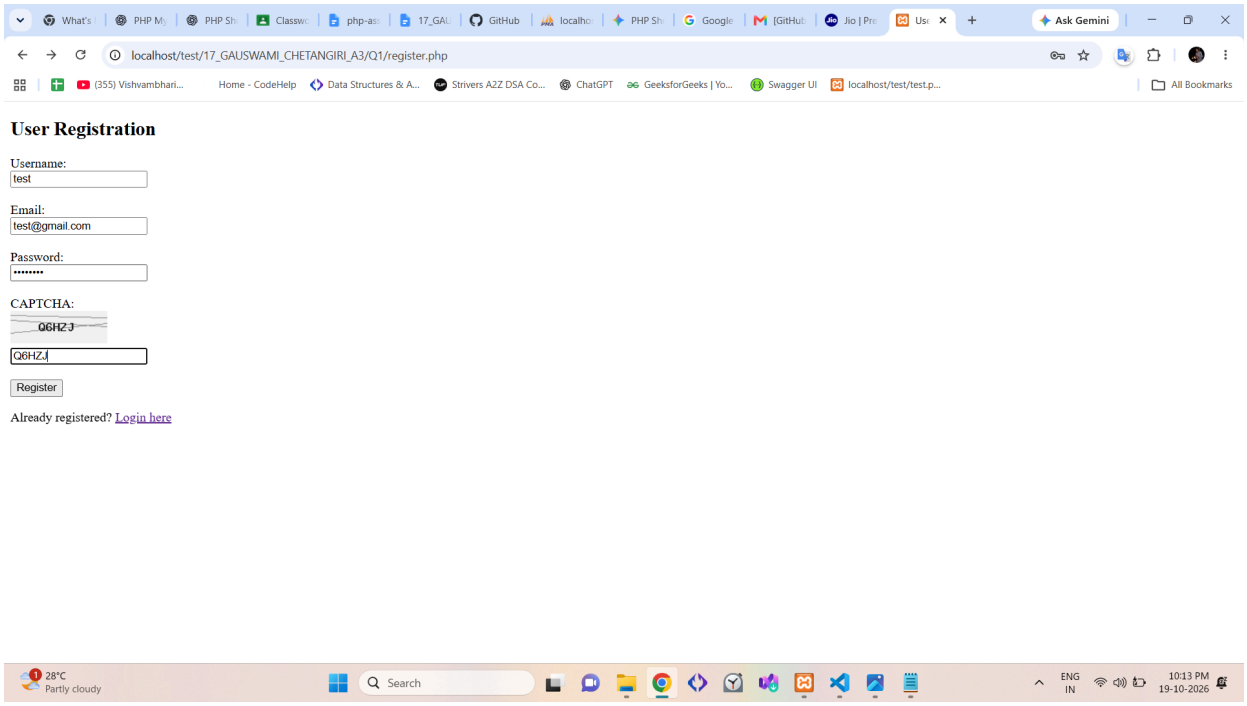
<tr><th>ID</th><th>Picture</th><th>Name</th><th>Category</th><th>Price</th
><th>Action</th></tr>
        <?php while($p = $products->fetch_assoc()): ?>
            <tr>
                <td><?= $p['id'] ?></td>
```

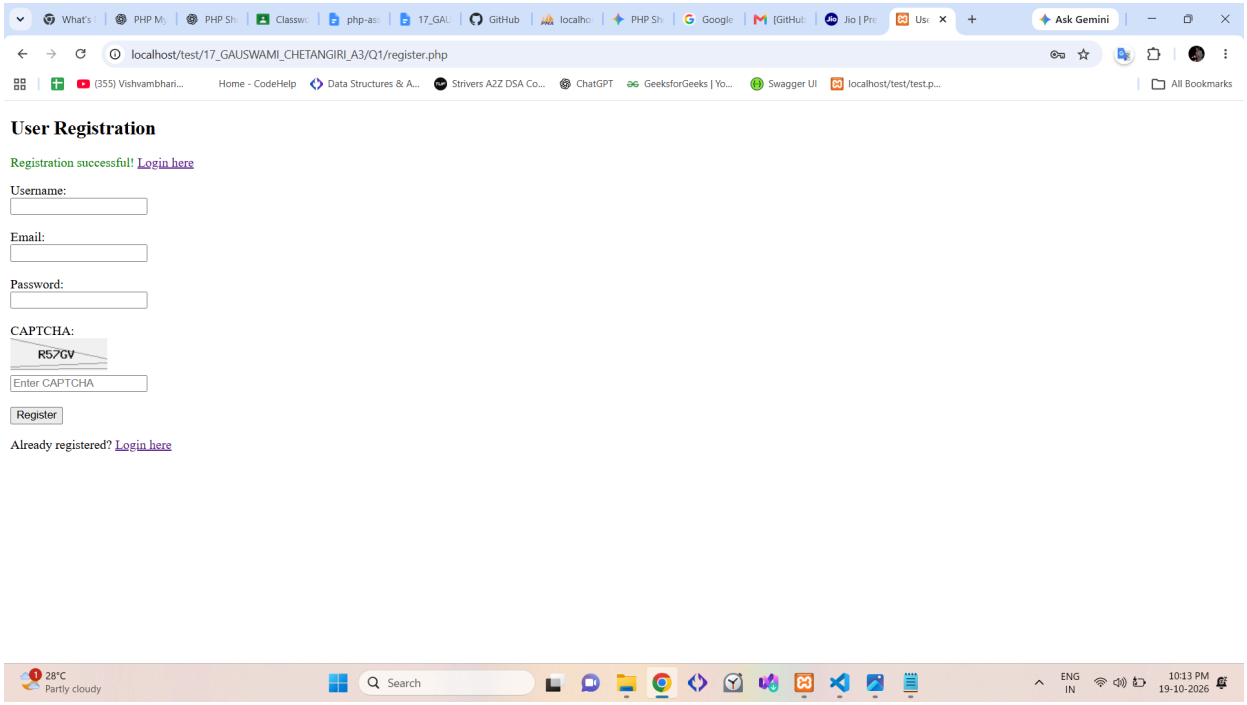
```
                <td><img src="<?= $p['image'] ?>" width="80" height="80"
style="object-fit:cover;"></td>
                <td><?= htmlspecialchars($p['name']) ?></td>
                <td><?= htmlspecialchars($p['category_name']) ?></td>
                <td>$<?= number_format($p['price'], 2) ?></td>
                <td><a href="?delete=<?= $p['id'] ?>" onclick="return
confirm('Delete this product?')">Delete</a></td>
            </tr>
        <?php endwhile; ?>
    </table>
</body>
</html>
```

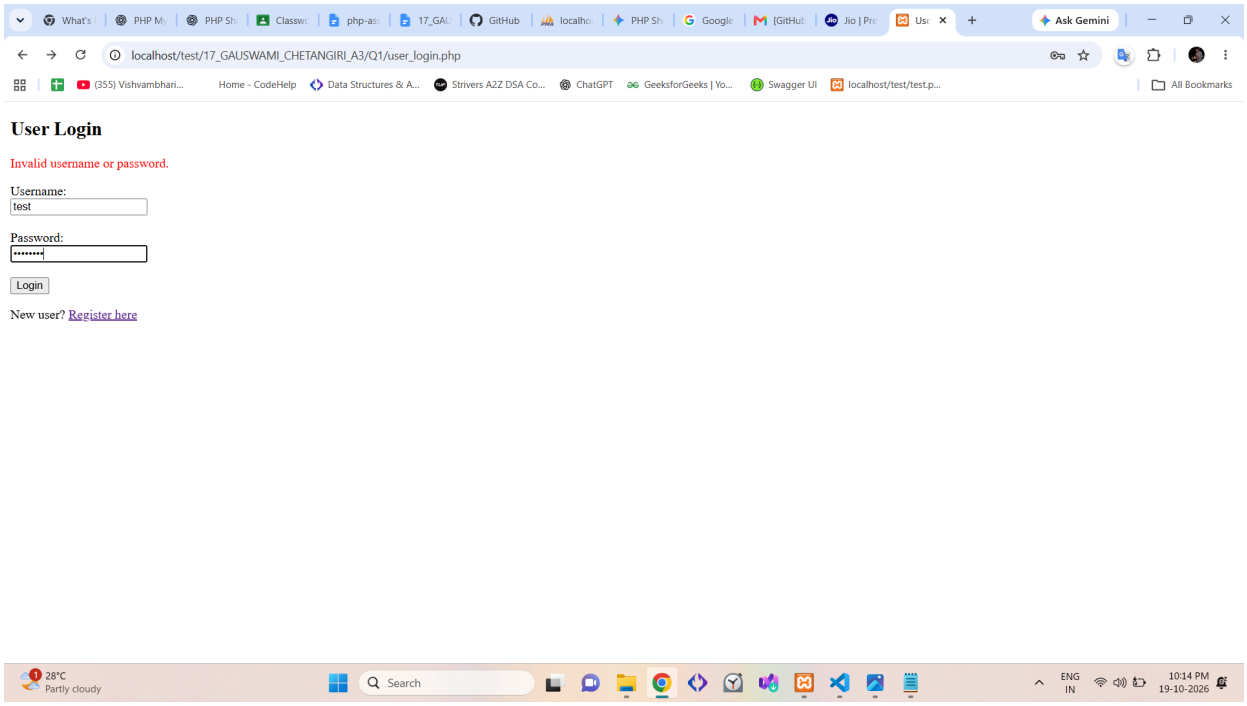
Logout

```
<?php
session_start();
session_unset();
session_destroy();
header("Location: admin_login.php");
exit;
?>
```

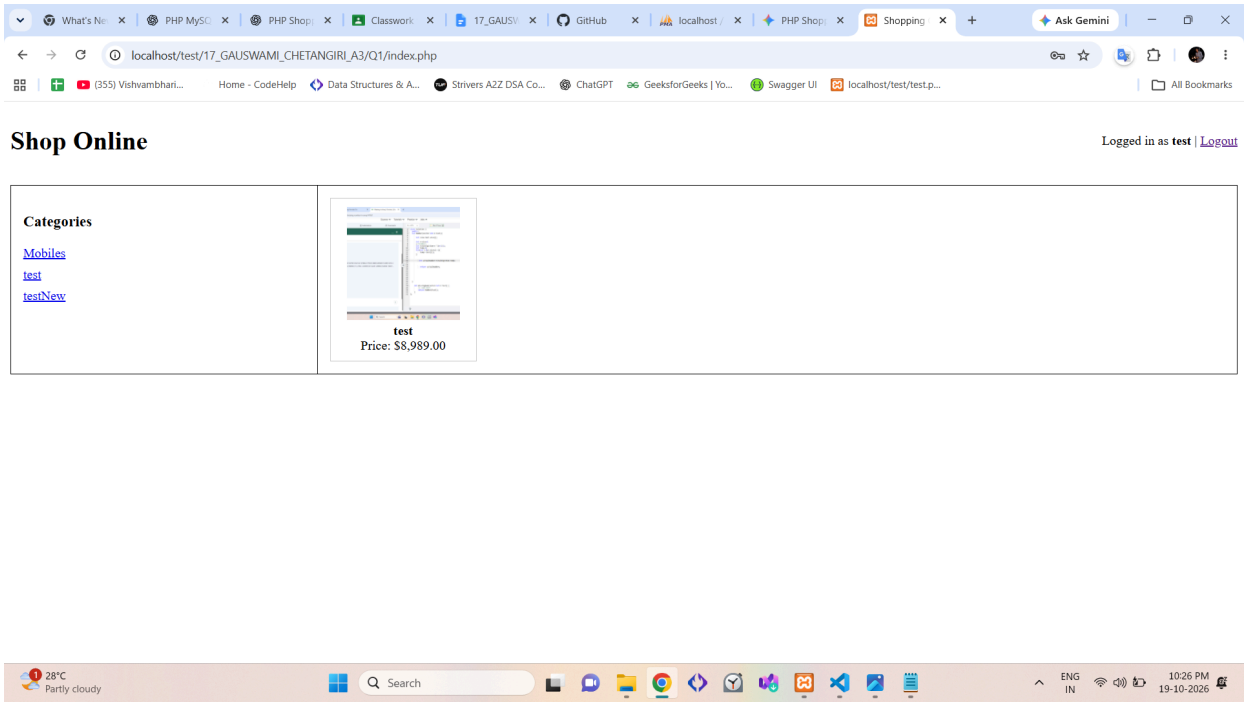
output=
Users


localhost/test/17_GAUSWAMI_CHETANGIRI_A3/Q1/register.php
User Registration
Username:
test
Email:
test@gmail.com
Password:
CAPTCHA:
Q6HZJ
Q6HZJ
Register
Already registered? Login here


localhost/test/17_GAUSWAMI_CHETANGIRI_A3/Q1/register.php
User Registration
Registration successful! Login here
Username:
Email:
Password:
CAPTCHA:
R57GV
Enter CAPTCHA
Register
Already registered? Login here



User with



Captch.php

```php
<?php
session_start();
```

```
$captcha_code = substr(str_shuffle("ABCDEFGHJKLMNPQRSTUVWXYZ23456789"), 0,
5);
$_SESSION['captcha'] = $captcha_code;

$img = imagecreatetruecolor(120, 40);
$bg = imagecolorallocate($img, 240, 240, 240);
$fg = imagecolorallocate($img, 20, 20, 20);
$line = imagecolorallocate($img, 150, 150, 150);

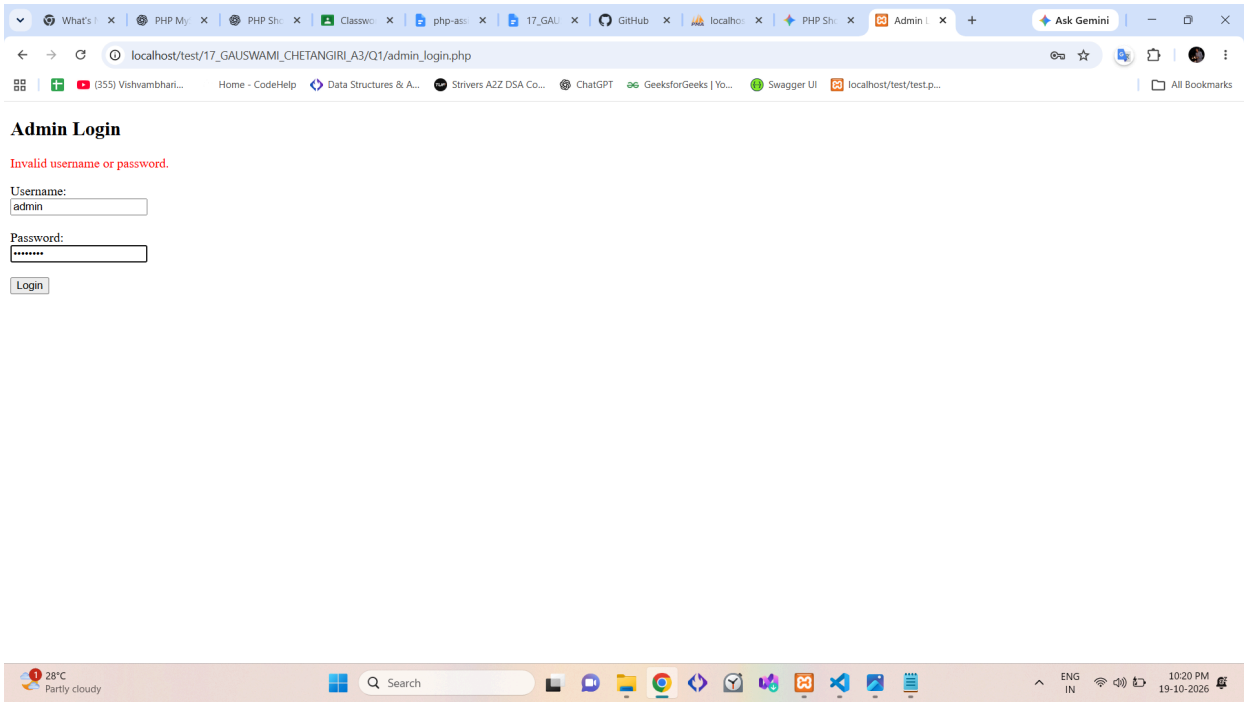
imagefilledrectangle($img, 0, 0, 120, 40, $bg);

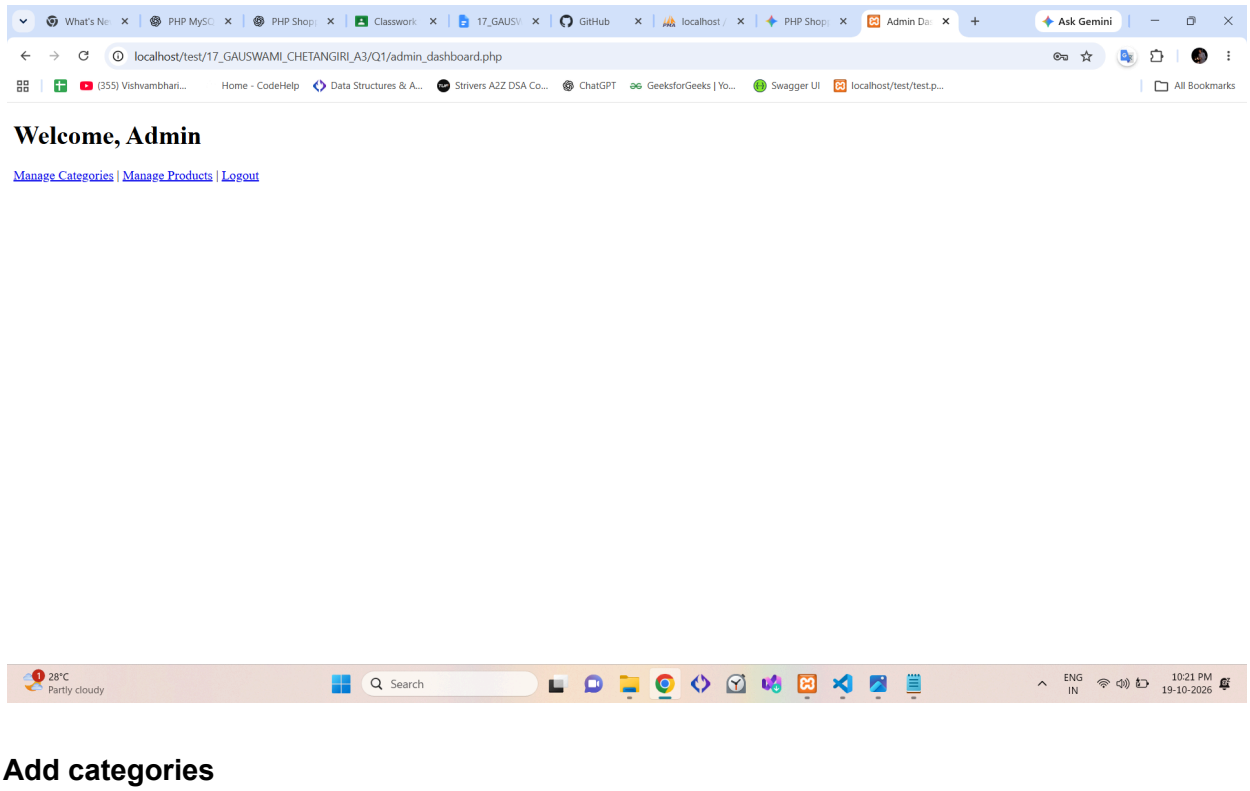
 for ($i = 0; $i < 3; $i++) {
    imageline($img, 0, rand(0, 40), 120, rand(0, 40), $line);
}

imagestring($img, 5, 35, 12, $captcha_code, $fg);

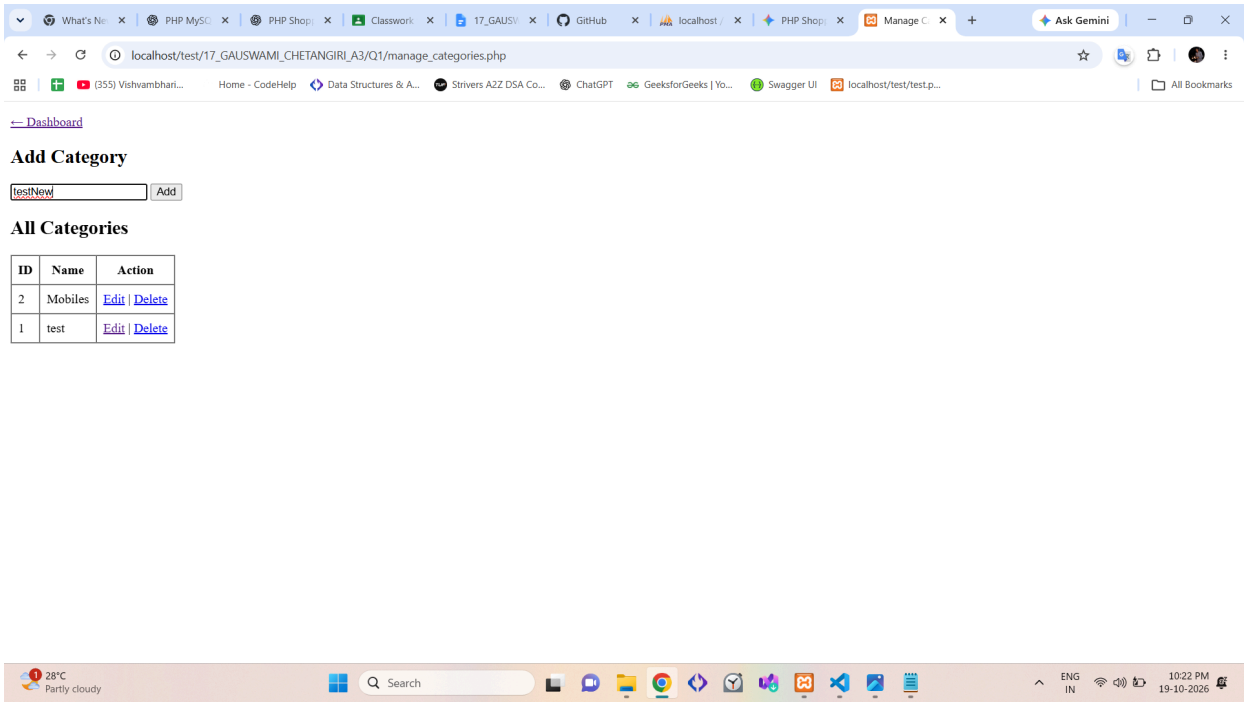
header("Content-Type: image/png");
imagepng($img);
imagedestroy($img);
?>
```

Admin

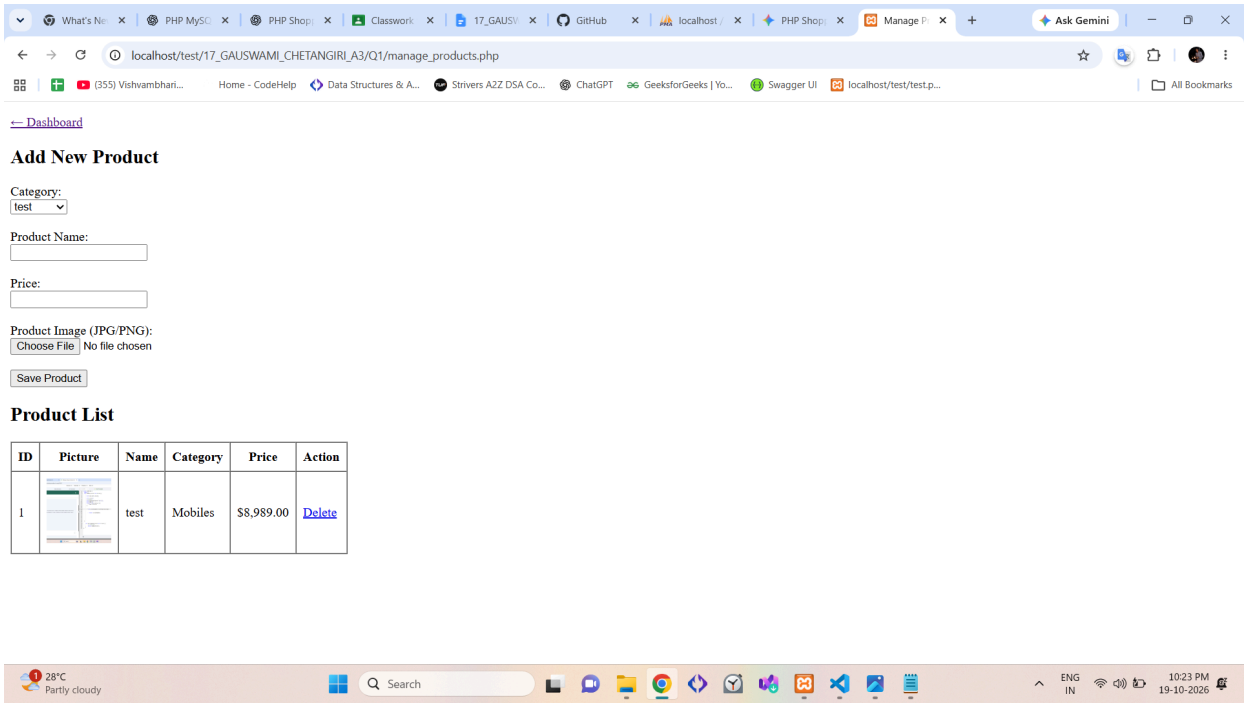




Add categories



Product Manage



Question 2: Read students.xml and store data in MySQL

Students.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>
<students>
    <student>
        <id>1</id>
        <name>Chetan</name>
        <email>chetan@gmail.com</email>
        <course>MSc ICT</course>
    </student>
    <student>
        <id>2</id>
        <name>Rahul</name>
        <email>rahul@gmail.com</email>
        <course>BCA</course>
    </student>
    <student>
        <id>3</id>
        <name>meet</name>
        <email>meet@gmail.com</email>
        <course>BSc IT</course>
    </student>
</students>
```

xml_to_mysql.php

```php
<?php
$conn = new mysqli("localhost", "root", "root", "datest");

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$xml = simplexml_load_file("students.xml");

if ($xml === false) {
    die("Unable to read students.xml");
}

$stmt = $conn->prepare(
    "INSERT INTO students (id, name, email, course)
     VALUES (?, ?, ?, ?)
     ON DUPLICATE KEY UPDATE
     name = VALUES(name),
     email = VALUES(email),
     course = VALUES(course)"
);

$stmt->bind_param("isss", $id, $name, $email, $course);

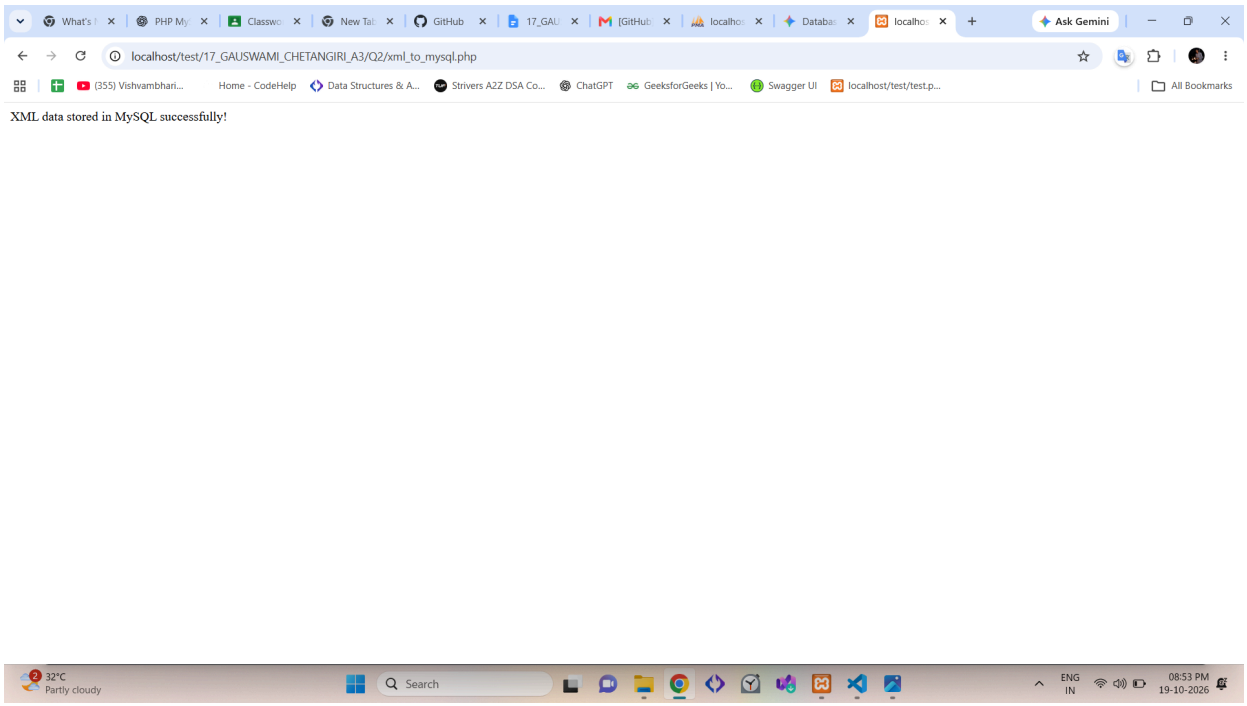
foreach ($xml->student as $student) {
    $id = (int)$student->id;
    $name = (string)$student->name;
    $email = (string)$student->email;
    $course = (string)$student->course;

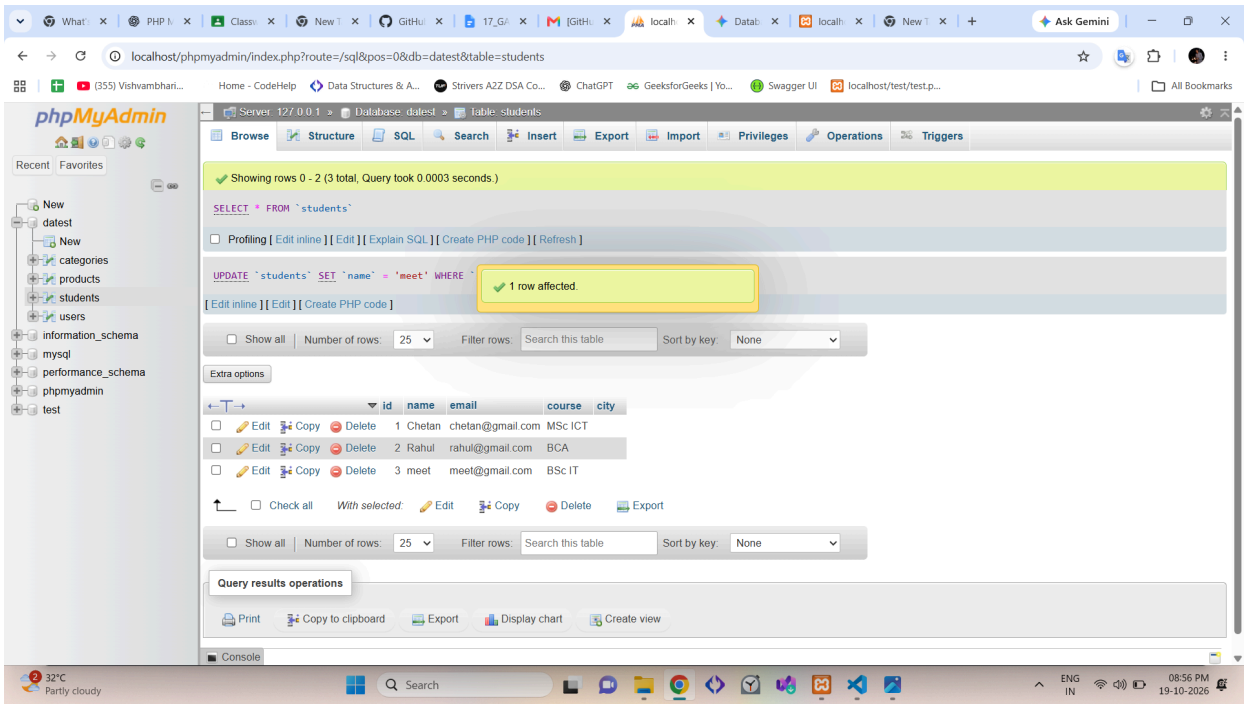
    $stmt->execute();
}

echo "XML data stored in MySQL successfully!";

$stmt->close();
$conn->close();
?>
```

Output


localhost/test/17_GAUSWAMI_CHETANGIRI_A3/Q2/xml_to_mysql.php
XML data stored in MySQL successfully!


localhost/phpmyadmin/index.php?route=/sql&pos=0&db=datest&table=students
phpMyAdmin
Server: 127.0.0.1 » Database: datest » Table: students
Browse
Structure
SQL
Search
Insert
Export
Import
Privileges
Operations
Triggers
Showing rows 0 - 2 (3 total, Query took 0.0003 seconds.)
SELECT * FROM `students`
1 row affected.
id
name
email
course
city
1 Chetan chetan@gmail.com MSc ICT
2 Rahul rahul@gmail.com BCA
3 meet meet@gmail.com BSc IT
Query results operations

Question 3: Read data from MySQL and store it in students.xml

get.php

```
<?php
$conn = new mysqli("localhost", "root", "root", "datest");

if ($conn->connect_error) {
    die("Connection : " . $conn->connect_error);
}

$result = $conn->query("SELECT id, name, email, course FROM students");

$xml = new SimpleXMLElement('<?xml version="1.0"
encoding="UTF-8"?><students/>');

while ($row = $result->fetch_assoc()) {
    $student = $xml->addChild("student");

    $student->addChild("id", $row["id"]);
    $student->addChild("name", htmlspecialchars($row["name"], ENT_XML1,
"UTF-8"));
    $student->addChild("email", htmlspecialchars($row["email"], ENT_XML1,
"UTF-8"));
    $student->addChild("course", htmlspecialchars($row["course"],
ENT_XML1, "UTF-8"));
}

$xml->asXML("students.xml");
```

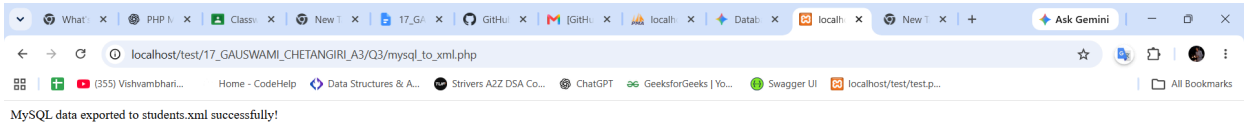
```php
echo "MySQL data exported to students.xml successfully!";

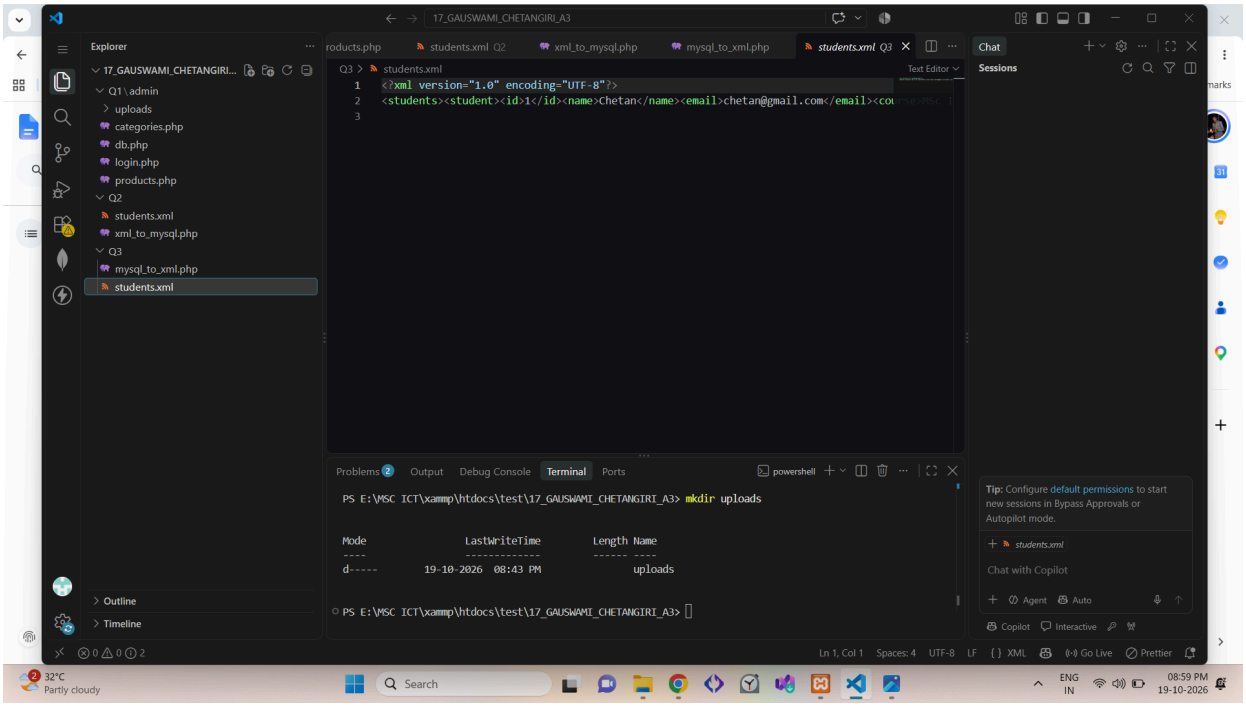
$conn->close();
?>
```

students.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>
<students><student><id>1</id><name>Chetan</name><email>chetan@gmail.com</email><course>MSc
ICT</course></student><student><id>2</id><name>Rahul</name><email>rahul@gmail.com</email><course>BCA</course></student><student><id>3</id><name>meet</name><email>meet@gmail.com</email><course>BSc
IT</course></student></students>
```

OUTPUT:





Question 4: List files in a directory using PHP

List_files.php

```php
<?php
$directory = "uploads";

if (!is_dir($directory)) {
    die("Directory does not exist!");
}

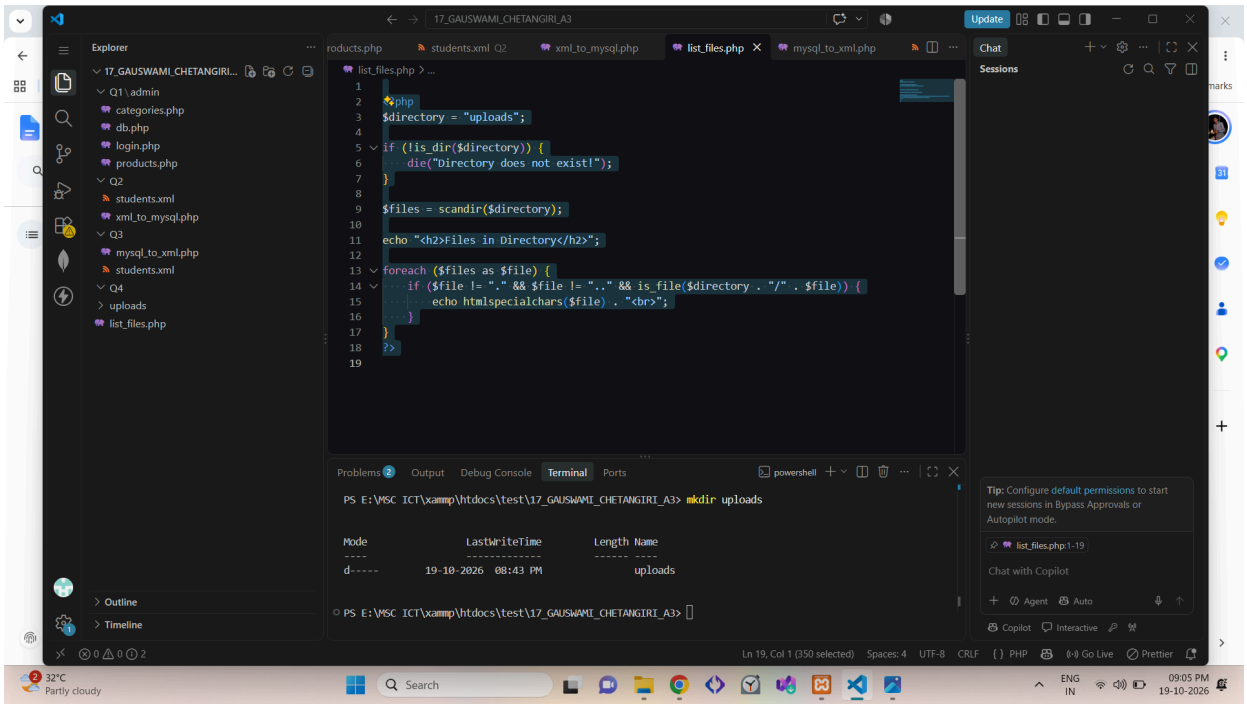
$files = scandir($directory);

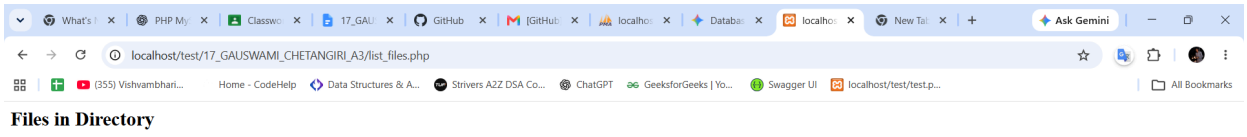
echo "<h2>Files in Directory</h2>";

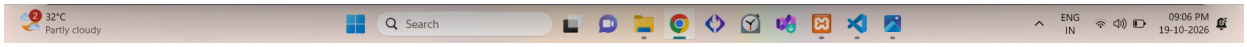
foreach ($files as $file) {
```

```
    if ($file != "." && $file != ".." && is_file($directory . "/" .
$file)) {
        echo htmlspecialchars($file) . "<br>";
    }
}
?>
```

Output







Question 5: Call an Express REST API in PHP

app.js

```
const express=require("express");
const session=require("express-session");
const mongoose=require("mongoose");
const emp=require("./models/emp");
const bcrypt=require("bcryptjs");
const app=express();

app.use(express.json());
app.use(express.urlencoded({"extended":true}));
app.use(session({
    secret:"this is secrete key",
    resave:false,
```

```
    saveUninitialized:false,
    cookie:{ maxAge:1000*60*60}

}));
app.set("view engine","ejs");

mongoose.connect("mongodb://localhost:27017/testDone").then(()=>{console.l
og("database connecte")});



app.get("/students", (req, res) => {
    const students = [
        { id: 1, name: "test", course: "MSc ICT" },
        { id: 2, name: "test", course: "BCA" },
        { id: 3, name: "test", course: "BSc IT" }
    ];

    res.json(students);
});



app.get("/",async (req,res)=>{


    const respose= await
fetch("http://localhost/test/17_GAUSWAMI_CHETANGIRI_A3/Q6/students_api.php
",{method:"GET"});
    const data=await respose.json();

    res.status(200).json(data);
});




app.listen(9090,()=>{
```

```
    console.log("url is  localhost:9090")
});
```

Call_express.php

```
<?php
$url = "http://localhost:3000/students";

$ch = curl_init($url);

curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
curl_setopt($ch, CURLOPT_TIMEOUT, 10);

$response = curl_exec($ch);
$status = curl_getinfo($ch, CURLINFO_HTTP_CODE);

if ($response === false) {
    die("API Error: " . curl_error($ch));
}

curl_close($ch);

if ($status != 200) {
    die("API returned HTTP status: " . $status);
}

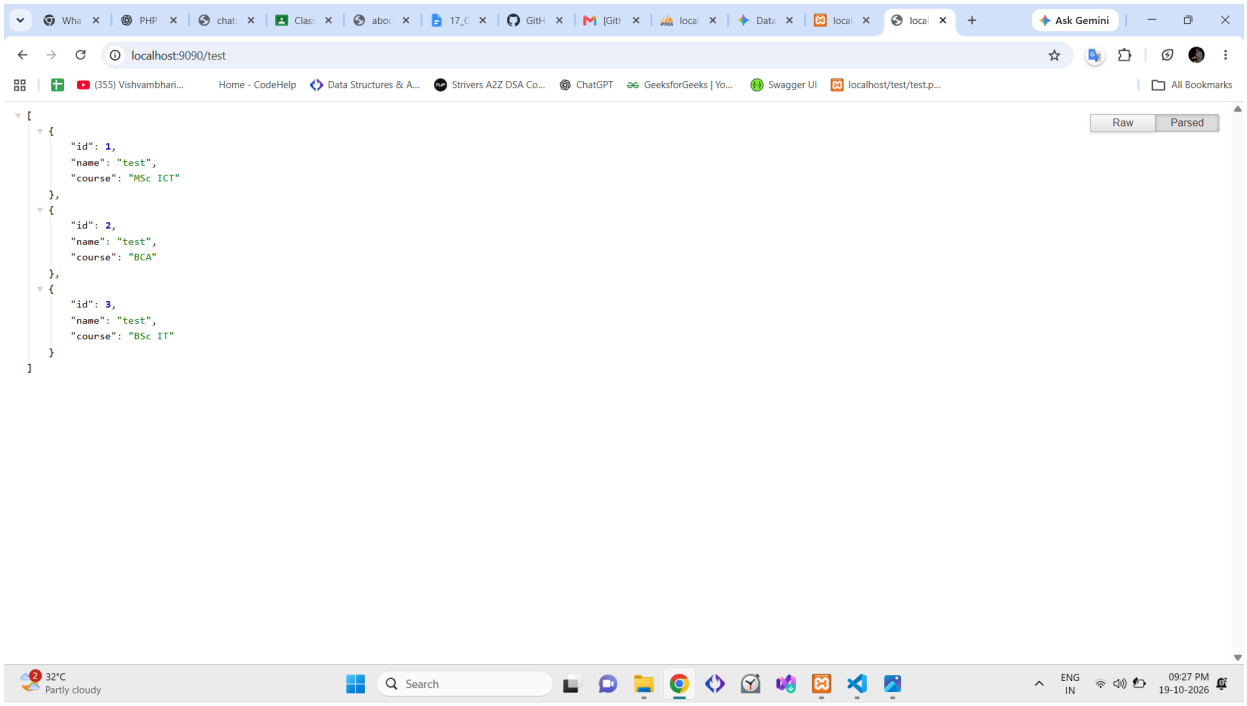
$students = json_decode($response, true);

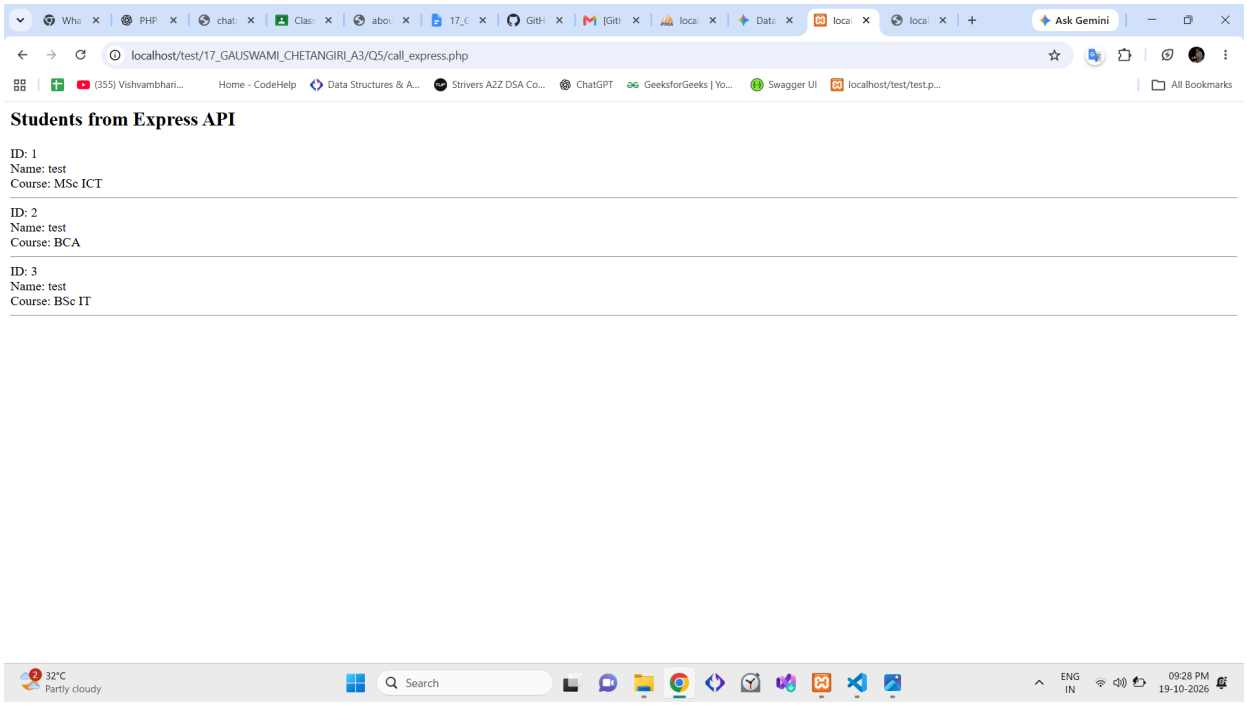
echo "<h2>Students from Express API</h2>";

foreach ($students as $student) {
    echo "ID: " . (int)$student["id"] . "<br>";
    echo "Name: " . htmlspecialchars($student["name"]) . "<br>";
    echo "Course: " . htmlspecialchars($student["course"]) . "<br><hr>";
```

```
}
?>
```

Output:


localhost:9090/test
Raw
Parsed
[
{
"id": 1,
"name": "test",
"course": "MSc ICT"
},
{
"id": 2,
"name": "test",
"course": "BCA"
},
{
"id": 3,
"name": "test",
"course": "BSc IT"
}
]
32°C
Partly cloudy
Search
ENG
IN
09:27 PM
19-10-2026



Question 6: Call a PHP REST API in Express.js

students_api.php

```php
<?php
header("Content-Type: application/json; charset=UTF-8");

$students = [
    ["id" => 1, "name" => "test", "course" => "MSc ICT"],
    ["id" => 2, "name" => "test", "course" => "BCA"],
    ["id" => 3, "name" => "test", "course" => "BSc IT"]
];

echo json_encode($students);
?>
```

app.js

```
const express=require("express");
const session=require("express-session");
const mongoose=require("mongoose");
const emp=require("./models/emp");
const bcrypt=require("bcryptjs");
const app=express();

app.use(express.json());
app.use(express.urlencoded({"extended":true}));
app.use(session({
    secret:"this is secrete key",
    resave:false,
    saveUninitialized:false,
    cookie:{ maxAge:1000*60*60}

}));
app.set("view engine","ejs");

mongoose.connect("mongodb://localhost:27017/testDone").then(()=>{console.l
og("database connecte")});









app.get("/",async (req,res)=>{


    const respose= await
fetch("http://localhost/test/17_GAUSWAMI_CHETANGIRI_A3/Q6/students_api.php
",{method:"GET"});
```

```
    const data=await respose.json();

    res.status(200).json(data);
});

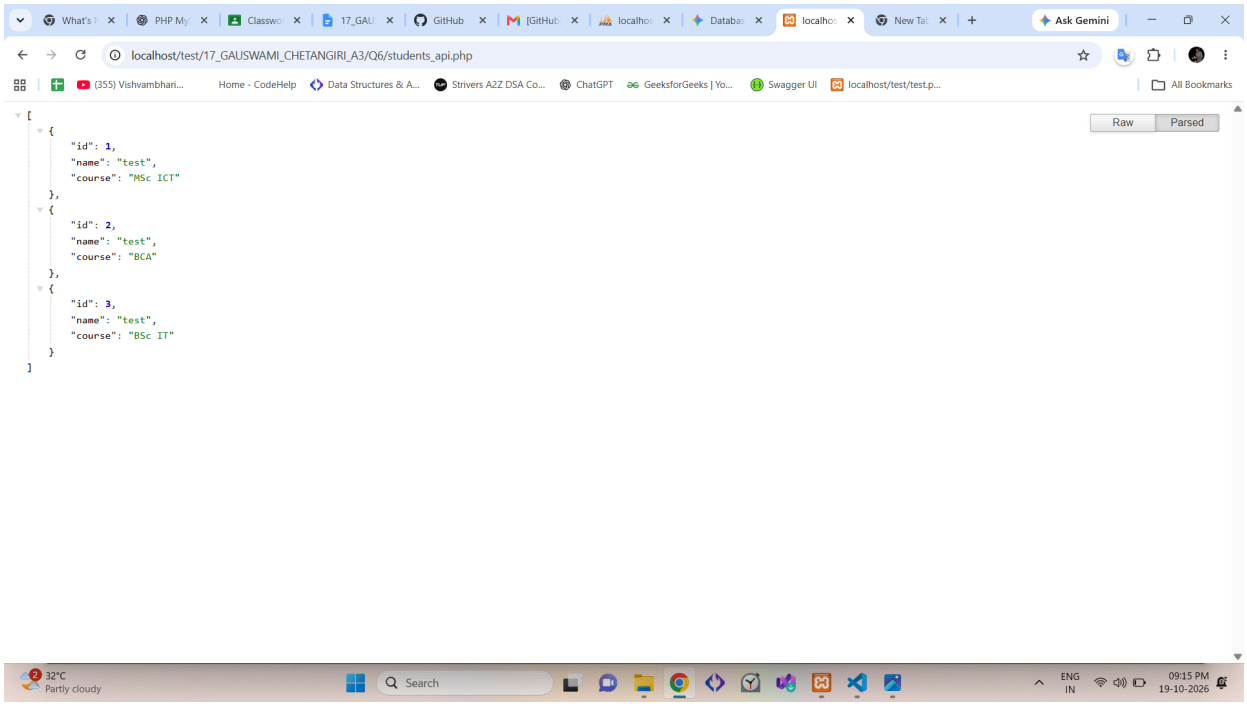



app.listen(9090,()=>{

    console.log("url is  localhost:9090")
});
```

Output:

Php output



Node js Output

